

Principal Motor Pathways

- **Corticospinal (pyramidal) system**- originates in the motor cortex, mediates voluntary movement and integrates skilled complicated movements through the stimulation and inhibition of certain muscle actions

Consists of both upper and lower motor neurons

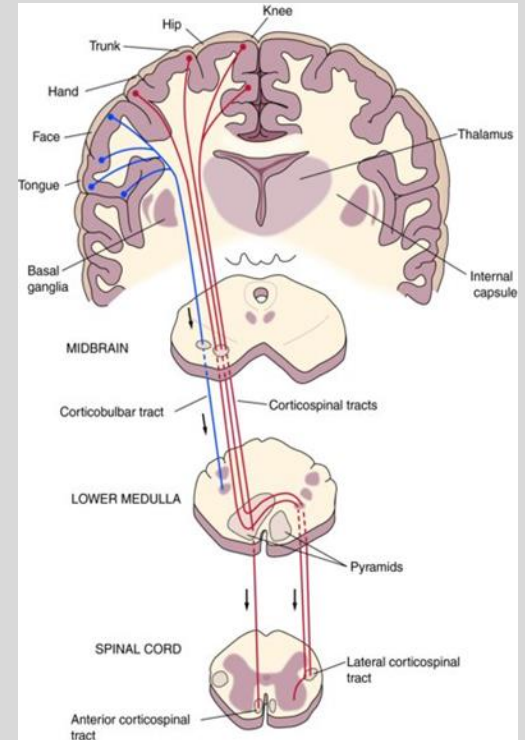
Synapse on the LMN in the spinal cord that mediate movement

Most fibers will cross over to contralateral side at the level of the medulla and spinal cord

(Tracts that synapse in the brainstem with the cranial nerve motor nuclei are the **corticobulbar tracts**.)

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Muscle Atrophy

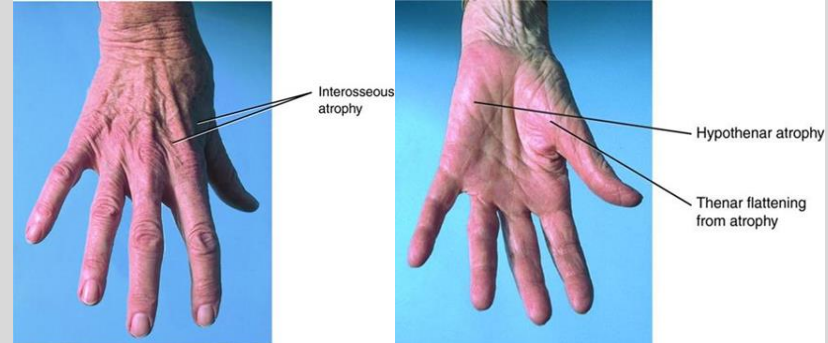
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- 44 yo- no atrophy



84 yo- with atrophy



Muscle Strength Scale

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Scale for muscle strength

- 0—No muscular contraction detected
- 1—A barely detectable flicker or trace of contraction
- 2—Active movement of the body part with gravity eliminated
- 3—Active movement against gravity
- 4—Active movement against gravity and some resistance
- 5—Active movement against full resistance without evident fatigue. (normal strength)

Source: Medical Research Council. Aids to the examination of the peripheral nervous system. London: Bailliere Tindall, 1986.

Documentation of Muscle Strength- 1/5, 2/5, 3/5, 4/5, 5/5

Upper Extremities- MS- Flexion/Extension

Flexion

Spinal nerve root innervations: C5, C6

- Patient is instructed to flex their arms at the elbow
- Examiner will instruct the patient to pull towards themselves (flex) as the examiner provides resistance

Extension

Spinal root innervations: C6, C7, C8

- Patient is instructed to flex their arms at the elbow
- Examiner will instruct the patient to push against the
- examiner's resistance.



Upper Extremities-MS- Abduction

Shoulder

Spinal nerve root innervations: C5, C6,
Axillary nerve

- Ask the patient to raise their arm from the side to shoulder level
- The examiner presses down firmly on the patient's upper arm with the shoulder abducted
- Instruct the patient to abduct against resistance



Wrist MS – Flexion/Extension

Flexion

- Spinal nerve root innervations: C6, C7, Median Nerve
- Instruct the patient to make a fist
- The patient flexes their wrist down against examiner's resistance pushing it upward.



Extension

Spinal nerve root innervations: C6, C7, C8, radial nerve

- Instruct the patient to make a fist
- The patient should push up (extend their wrist) to resist as the examiner pushes down on the fist



Finger MS- Adduction/Abduction

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Finger Adduction - MS

Spinal nerve root innervations: C7,C8, T1

- The patient is asked to squeeze two of the examiner's fingers as hard as they can and to not let go



Finger Abduction- MS

Spinal nerve root innervations: C8, T1, Ulnar nerve

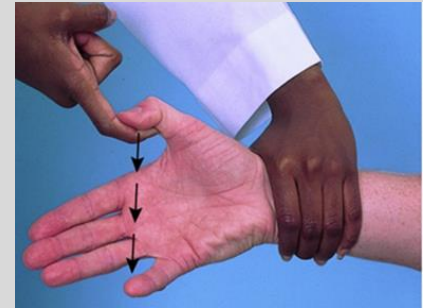
- Position the patients' hand with the palm facing down and fingers spread.
- Instruct the patient to prevent you from moving any fingers as you try to force them together



Opposition of the Thumb- MS

Spinal nerve root innervations:
C8, T1, median nerve

- Ask the patient to touch the tip of your finger with their thumb against your resistance



Lower Extremities –MS-Hip Flexion/Extension

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Flexion

Spinal nerve root innervations: L2, L3, L4

- Examiner's hand is on the anterior part of the patient's mid thigh
- Patient is instructed to raise their leg up against the examiner's resistance



Extension

Spinal nerve root innervations: S1

- Examiner's hand is on the posterior part of the patient's mid thigh
- Patient is instructed to push their thigh down against the examiners resistance



Lower Extremities- MS- Hip Adduction/Abduction

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Adduction

Spinal nerve root innervations: L2, L3, L4

- Examiner places their hands on the inside of the patient's knees
- Patient is instructed to bring their legs together against the examiner's resistance

Abduction

Spinal nerve root innervations: L4, L5, S1

- Examiner places their hands on the outside of the patient's knees
- Patient is instructed to spread their legs apart against the examiner's resistance



Lower Extremities- MS Knee Flexion/Extension

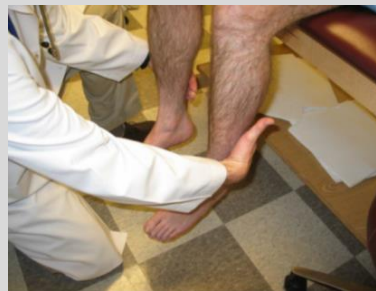
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Flexion

Spinal nerve root innervations: L5, S1, S2

- Patient is instructed to pull (flex) their legs against the examiner's resistance



Extension

Spinal nerve root innervations: L2, L3, L4

- Patient is instructed to push (extend) their legs against the examiner's resistance



Lower Extremities- MS- Ankle

Dorsiflexion

Spinal nerve root innervations: L4, L5

- Patient is instructed to bring toes upward against resistance from the examiner



Plantar Flexion

Spinal nerve root innervations: S1

- Patient is instructed to push foot downward against resistance from the examiner



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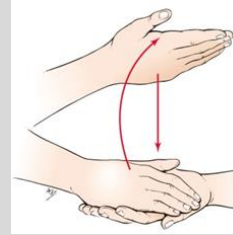
Rapid Alternating Movements

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- **Arms (using hands)**

Rapid alternation of pronation of one hand with supination on the other hand



- **Fingers**

Touch the thumb to each finger quickly



- **Legs**

Tap the floor or examiner's hand with the ball of the foot quickly



Observe for the speed, rhythm and smoothness of the movements

Point to Point Movements- Arms/Legs

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Arms

Finger to Nose

- Patient alternates touching the Examiner's index finger and then their own nose.
- Examiner changes the position of their finger.
- Observe for smoothness and accuracy
- Dysmetria- overshoot the mark initially and then reach it



Legs

Legs-Heel-Shin

- Patient places one heel on the opposite knee and will slide it down the shin of the other leg
- Observe for smoothness and accuracy



Gait- Evaluation

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Walk across the room/down the hall

- Observe posture, balance, the swinging of the arms, and leg movements
- Normal- intact balance, arms swing symmetrically at the sides and turns are smooth
- Tandem Walking- walk in straight line using heel to toe

May indicate ataxia not otherwise seen

- Walk on toes then heels
- Hop in place
- Shallow knee bend- on one leg then the other



Reflex Hammers

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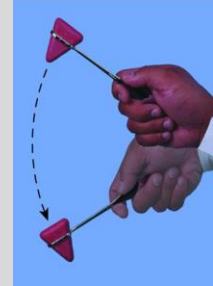
Deep Tendon Reflex

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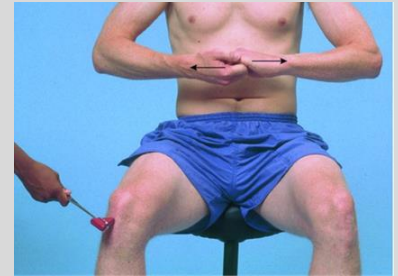
Procedure

- Patient should be encouraged to relax
- Reflex hammer should be held loosely between your thumb and index finger so that it swings freely in an arc within the limits of your palm and fingers
- Note speed, force and amplitude of the reflex response and grade it using the grading scale



Use techniques listed below for patients who have symmetrically diminished or absent reflexes

- For arm reflexes- ask patient to clench their teeth or squeeze their knees together
- For leg reflexes- ask patient to lock their fingers and pull against the other (Jendrassik's maneuver)



Deep Tendon Reflexes

Biceps- Spinal nerve roots- C5,C6

- Examiner partially flexes the patient's arm
- Strike the biceps tendon (alternative- examiner places their thumb or finger on the biceps tendon and then strike their finger)
- Assess bilaterally

Brachioradialis- Spinal nerve roots- C5,C6

- Patient's hand is in a relaxed position-on lap, table or supported by the examiner's arm.
- Examiner strikes the radius about 1 or 2 inches above the wrist with the reflex hammer
- Assess bilaterally



Deep Tendon Reflexes

Triceps- Spinal nerve roots- C6,C7

- Examiner flexes the patient's arm at the elbow and supports the upper arm. Ask the patient to let their arm go limp.
- Strike the triceps tendon with the reflex hammer
- Assess bilaterally



Quadriceps (Patellar)-Spinal nerve roots L2, L3,L4

- Have patient sit on the edge of the table with legs flexed
- Strike the patellar tendon using the reflex hammer
- Assess bilaterally



Deep Tendon Reflexes

Achilles (Ankle)

Spinal nerve root (Primarily S1)

- Have patient remove shoes and socks
- Examiner dorsiflex the patient's foot at the ankle
- Achilles tendon is struck with the reflex hammer
- Assess bilaterally



Ankle Clonus

- Clonus is tested when the reflexes appear to be hyperactive.
- Knee is supported in flexed position
- Dorsiflex and plantar flex the foot to relax it
- Sharply dorsiflex the foot
- Response- rhythmic oscillations
- Must be present in order to grade a stretch muscle reflex 4+

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Special Techniques

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- Plantar Response

Examiner strokes the lateral aspect of the sole of the foot from the heel to the ball of the foot



Plantar Flexion- Normal



Plantar Dorsiflexion-
Babinski Response- CNS lesion
affecting the Corticospinal Tract

Special Technique

Hoffmann's Test

Test for upper motor neuron lesion- Above T1

- Flick the nail of the middle finger
- If the thumb and 2nd finger flex, this is a positive Hoffman's Test. (See figure 2.70)
- The patient has neurological symptoms due to an upper motor neuron lesion and not to a radiculopathy or peripheral nerve lesion

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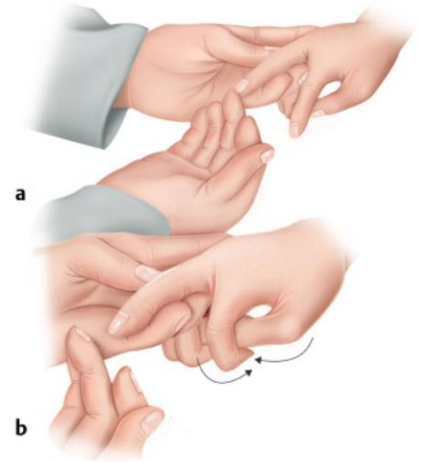


Fig. 2.70 (a) Hoffmann's test. (b) Positive Hoffmann's test with finger and thumb flexion after flicking the middle fingernail.

Special Techniques

- **Romberg Test**

- Test of position sense
- Feet together with eyes open
- If there is no loss of balance continue
- Close both eyes for 30-60 seconds

Positive Romberg Test

- Loss of balance when eyes are closed
(Dorsal Column disease)

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Special Techniques

- **Pronator Drift Test**

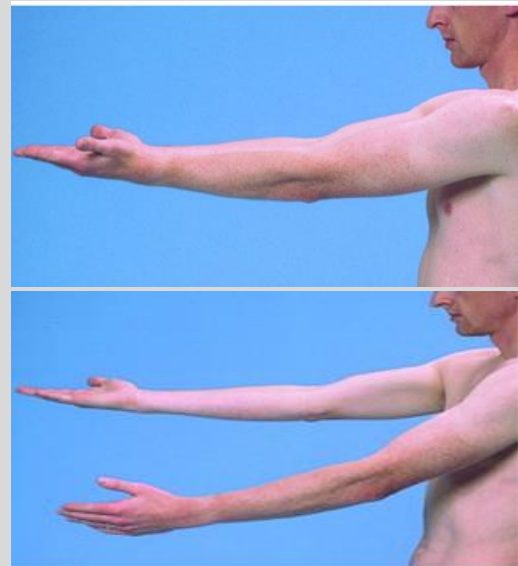
- Eyes closed and both arms straight forward with palm up for 20-30 seconds
- Maintain above position
- Examiner taps the arms briskly downward.

Normal response- arms return to original position

Pronator Drift- one forearm and palm turn inward and down (corticospinal tract lesion in contralateral hemisphere)

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Documentation of a Normal Neurological Examination

Neuro-

Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension

- CN I not tested CN II-XII grossly intact
- Motor strength is 5/5 in all extremities, good muscle bulk and tone
- Rapid alternating movements and finger to nose are intact
- Gait is normal, no evidence of ataxia
- Negative Romberg and Pronator Drift
- Deep Tendon Reflexes 2+ bilaterally, plantar reflexes downgoing or plantar flexion bilaterally